Sanity-LLM: Enhancing Mental Health Chatbots with Retrieval-Augmented Generation (RAG)

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2025 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS) | 979-8-3315-7930-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICIMCIS68501.2025.11327020

***Abstract*— This study introduces Sanity-LLM, an advanced Retrieval-Augmented Generation (RAG) framework for mental health chatbots that aims to produce responses that are accurate, empathetic, and safe. The proposed architecture integrates hybrid semantic–keyword search, step-back questioning, and a reranking mechanism to improve the relevance and grounding of generated responses. The knowledge base was built from publicly available mental health dialogues and Q&A data, embedded and indexed using FAISS. Evaluation using RAGAS metrics demonstrates notable performance improvements, achieving 0.9391 context precision, 0.7077 context recall, 0.8754 faithfulness, and 0.6297 answer correctness, surpassing naïve RAG baselines. Ablation results further confirm that each module: hybrid retrieval, step-back prompting, and reranking contributes significantly to the overall system effectiveness. While not a substitute for professional care, Sanity-LLM offers a scalable and accessible first-line support tool for encouraging help-seeking behavior. Future work will focus on broadening verified knowledge sources and validating effectiveness through expert and user-based evaluation.**

Keywords—Advanced RAG, Mental Health Chatbot, Hybrid Search, FAISS, Large Language Model

I. INTRODUCTION

Mental health issues have become a growing global concern, affecting individuals across diverse demographics and socioeconomic backgrounds. According to the World Health Organization (WHO), mental health disorders contribute substantially to the global burden of disease, with affected individuals having a life expectancy that is 10 to 20 years shorter than the general population [1]. Despite this alarming reality, many people still lack access to timely, affordable, and effective mental health support services.

Although awareness of mental health has increased in recent years, societal stereotypes and stigma continue to influence how it is perceived. Many individuals refrain from seeking professional help due to fears of being labeled as "weak" or "unstable" [2], while others may suppress their struggles in cultures that value emotional restraint and resilience. Such misconceptions hinder access to proper mental health support, often resulting in misdiagnosis or inadequate treatment. To help address this challenge, this research explores the use of Artificial Intelligence (AI) to provide more accessible mental health consultation [2].

The emergence of large language models (LLMs) has transformed natural language processing, enabling more advanced human-computer interaction and deeper semantic understanding of text [3]. However, applying general-purpose LLMs in mental health contexts poses significant challenges. These models often lack domain-specific knowledge essential for accurately understanding and responding to mental health concerns. Without enrichment from reliable, expert-curated datasets, LLMs risk producing responses that may be inaccurate, insensitive, or even potentially harmful.

Previous studies have investigated the application of machine learning techniques for mental health detection. For instance, a systematic review [4] examined various machine learning techniques employed in Online Social Networks (OSNs) to identify mental health conditions, highlighting both the potential and the limitations of these approaches. However, the integration of domain-specific enhancements into LLMs remains underexplored, creating a gap in their effectiveness and applicability for therapeutic or supportive interventions.

Retrieval Augmented Generation (RAG) has emerged as a promising solution to the limitations of traditional chatbots. By integrating external knowledge sources with LLMs, RAG enables the generation of responses grounded in up-to-date medical information and tailored with greater empathy. This approach also mitigates common issues associated with LLMs, such as hallucinations and outdated knowledge. Furthermore, RAG facilitates automation in the healthcare domain, where chatbots can assist professionals by addressing general patient inquiries. As a result, it has the potential to reduce patient waiting times and enhance the overall efficiency of healthcare service delivery.

Previous studies have explored the use of RAG in various contexts related to mental health support. One common approach involves integrating RAG with question–answer (Q&A) datasets to enhance the ability of large language models (LLMs) to generate more empathetic and contextually appropriate responses, an aspect that is highly important in mental healthcare. Another line of research focuses on leveraging RAG to incorporate the latest findings and knowledge related to mental health treatment and interventions. Since LLMs often lack access to up-to-date or domain-specific information, the use of RAG becomes

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essential for improving their accuracy and reliability in generating informed responses.

This research focuses on enhancing LLMs with domain-specific mental health knowledge, aiming to bridge the gap between general linguistic capabilities and specialized therapeutic applications. By integrating curated knowledge bases, clinical guidelines, and expert feedback, the enhanced model is expected to provide more accurate, empathetic, and contextually appropriate support to individuals experiencing mental health challenges.

The expected outcomes of this study include sanity-LLM, an improved LLM capable of delivering more accurate and empathetic responses in mental health contexts, a framework for integrating domain-specific knowledge into LLMs that can be extended to other fields. By combining advancements in AI with a focus on mental health, this research aims to address the accessibility gap in psychological support, particularly in underserved communities, and contribute to a future where technology plays a pivotal role in promoting mental well-being.

II. Literature Review

In recent years, LLMs have been increasingly applied to solve real-world problems across diverse domains. Common approaches include integrating LLMs with RAG frameworks to enhance factual accuracy, fine-tuning models for domain specialization, and combining them with traditional deep learning techniques for better performance. These strategies demonstrate the versatility of RAG–LLM systems in various contexts.

RAG has been explored in multiple domains such as agriculture, education, and business, highlighting its broad applicability. In agriculture, RAG has been combined with hybrid deep learning models to identify medicinal plants for disease treatment [5], while other works developed RAG-based virtual assistants to support farmers in improving productivity [6,7]. In education, RAG has been utilized for quiz generation and virtual tutoring [8-10], demonstrating improvements in engagement and personalized learning. Similarly, in business and finance, RAG-assisted systems have been used for data aggregation, decision support, and financial analysis [11-13]

Focusing specifically on mental health applications, the architectures of RAG can generally be categorized into two types: Naive RAG and advanced RAG.

A. Naive RAG

Research on SentimentCareBot [14] combines RAG with sentiment analysis techniques to enhance conversational quality and chatbot responses. The retrieval process is based on a publicly available mental health counseling dataset. The integration of RAG and sentiment analysis demonstrated superior performance compared to baseline RAG models. However, this improvement comes with a trade-off in computational efficiency due to the additional sentiment re-ranking process. Another study utilized quantized LLMs combined with RAG [15] to analyze user documents before generating responses or diagnostic suggestions. Quantized LLMs offer advantages in terms of computational efficiency, allowing them to operate on limited hardware resources. However, their accuracy may decrease because of the lack of fine-tuning after model size reduction. Moreover, such a chatbot system also raises privacy concerns, particularly in the handling of sensitive user data, as there remains a potential risk of data misuse during model improvement.

Considering the trade-off between computational cost and accuracy, another study develops Medbot [16], which focuses on utilizing RAG without combining additional methods. The system retrieves data from PubMed sources and uses it to provide answers to users' medical inquiries, including potential diagnostic suggestions. However, the study provides limited details on chatbot responses from a professional evaluation perspective. Allowing the chatbot to generate diagnostic outputs directly raises concerns about the possibility of false diagnoses, especially given the inherent limitations of LLM accuracy. A similar approach was adopted in research on LLMs assisting sexual harassment victims [17], where the Llama-2 model was used to retrieve legal resources. With that method, their LLM managed to achieve an accuracy of 95%, demonstrating its ability to deliver sensitive and supportive assistance in trauma-related contexts. Marco et al. [18], in this recent research, proposed the RAG-LLM model based on ICD-11, a new classification system for physical and mental disorders. They introduced LLMind chat that utilizes Gemma 2 LLM. By leveraging the improved classification system, the model aims to better assist mental health professionals. The performance of their LLM was assessed by the DSM-5-TR Clinical cases manual to ensure that the responses from the chatbot were both credible and user-friendly.

B. Advanced RAG

Recently, more advanced research has been proposed. LLMs integrated with RAG are no longer limited to simple question-answering tasks but are now being utilized for a variety of purposes. In 2024, Qiming et al. [19] proposed an application called Soullmate, designed to address gaps in existing mental health applications. By combining LLMs, prompt engineering, and RAG, they developed a system capable of multiple functionalities, including question-answering, mental health assessment, and risk detection. Their model was evaluated using the DAIC-WOZ dataset, Reddit suicide posts, and other case studies. With GPT-4 + SMMR, they demonstrated an improvement in performance across these tasks. Another recent study by Ahsan et al. [20] introduced online reinforcement learning to enhance RAG utilization. Based on similarity scores, their approach dynamically improved model performance in real time. The reinforcement learning component determines whether to retrieve additional data or refine existing responses based on contextual similarity. Their dataset consisted of mental health issues collected from college student surveys. Through evaluations with GPT-4o, GPT-4o-mini, Gemini 1.5, and GPT-3.5, the proposed OnRL-RAG model outperformed all baselines in similarity to ground-truth responses, demonstrating its ability to generate more personalized and contextually relevant outputs.

Another study, [21] LLM-Therapist explores how fine-tuned LLMs can analyze patient data and generate responses based on the provided information. This approach enables more personalized and context-aware responses from the chatbot. The researchers evaluated their model using relevance and user satisfaction metrics, achieving strong results in both aspects. The application of RAG has also been extended to text-to-speech systems [22], where the study utilized a knowledge graph-based retrieval mechanism. The researchers compared three RAG architectures: Ensemble

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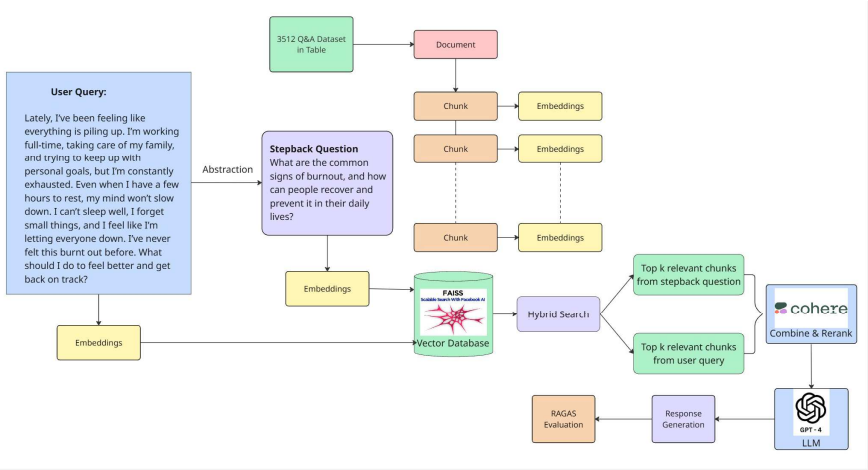


Fig. 1. System Architecture of Sanity-LLM with RAG based stepback question, Hybrid Search, and Reranking

retriever-based RAG, auto-merging retriever-based RAG, and naive RAG, and found that the ensemble retriever configuration achieved the best performance. Additionally, another advanced RAG implementation focused on enhancing patient diagnosis using models such as BioMistral7B and PubMedBert [23]. Evaluation using BLEU and ROUGE scores indicated that the chatbot produced high-quality responses, demonstrating strong potential for further development in medical and diagnostic applications.

III. Methodology

The system architecture for Sanity-LLM, as shown in Fig. 1, comprises seven key sections: dataset preparation, query optimization, hybrid search, reranking, LLM Selection & prompt engineering, test set creation, and lastly RAG evaluation.

A. Dataset Preparation

To prepare the data, a preprocessing pipeline was applied to normalize text by removing extra whitespace, correcting spacing around punctuation, and standardizing quotation marks. Entries with null or excessively short responses were removed, and duplicate *(Context, Response)* pairs were dropped while retaining multiple valid answers for the same context. The resulting dataset was then grouped so that each context could include several bullet-point responses, ensuring both cleanliness and diversity in the knowledge base.

Finally, the combined dataset was embedded and indexed using Facebook AI Similarity Search (FAISS) to enable dense vector retrieval. The *context* component was also tokenized and processed with BM25 to support hybrid search in later stages.

Since the data concerns chats and conversations between patients and doctors and may contain sensitive and even personal information, we have taken steps to anonymize the data to ensure the safety of the individuals who participated in the conversations by erasing any names, addresses, phone numbers/emails, or other personal details before utilizing the data in our system.

B. Query Optimization

For query optimization, user queries are processed using a stepback prompting technique. This approach transforms the original query into a simpler and more fundamental form, enabling the model to better understand the underlying intent of the question. By generating stepback questions, the system enhances the accuracy of document retrieval and ensures that the retrieved information is more closely aligned with the user's actual needs.

C. Hybrid Search

As illustrated in Fig.2, both the original query and the corresponding stepback question undergo a hybrid search on the FAISS Database to retrieve the most relevant document. This hybrid approach is one of the most widely adopted techniques in advanced RAG systems, as it combines keyword-based and semantic-based retrieval. On the keyword side, the query is matched against tokenized documents using the BM25 algorithm, which computes relevance scores based on keyword frequency and distribution. Simultaneously, on the semantic side, the query is embedded and compared with the embedded documents stored in FAISS, where cosine similarity is used to measure semantic closeness and generate semantic scores. To ensure comparability, both BM25 and semantic scores are normalized using Min–Max scaling, after which the scores are combined into a single fusion score. Finally, documents are ranked according to their fusion scores, and the top-N most relevant documents are selected for subsequent processing in the RAG system.

To determine the optimal weighting between the two retrieval methods, experiments were conducted across different weight ratios ranging from 0.1 to 0.9. The performance of each configuration was evaluated using RAGAS metrics, and the configuration yielding the best performance was selected. Experimental results showed that the optimal balance occurred when the BM25 weight was 0.2 and the FAISS weight was 0.8. This configuration achieved the best retrieval accuracy by balancing the strengths of keyword and semantic search.

D. Reranking

After retrieving the documents relevant to both the original query and the stepback question, the next step is to combine and rerank them to ens ure the most relevant information is prioritized. Before feeding the retrieved documents into the LLM, all candidate contexts are reranked based on their re levance to the user's original query. This reranking process was implemented using Cohere's Rerank API with the *rerank-*

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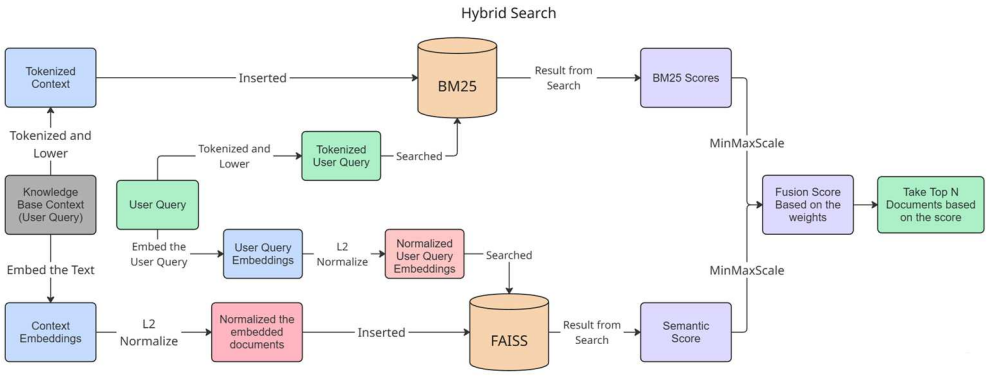


Fig. 2. Hybrid Search Process

english-v3.0 model, which assigns a semantic similarity score between each query–document pair. The retrieved contexts (top-10 from the hybrid search) were reranked according to these scores, and the top-5 highest-scoring contexts were selected and passed into the LLM for the final response generation. This configuration helps balance retrieval diversity with response precision while maintaining computational efficiency.

E. LLM Selection & Prompt Engineering

In this system, the Large Language Model (LLM) plays a crucial role. The embedding layer is primarily used during the retrieval phase of the RAG process. Before performing a search in the vector database, both the user's query and the stored documents are converted into embeddings to enable semantic similarity comparison. This embedding process is essential for achieving accurate semantic search. For embedding, we used the text-embedding-3-small model from OpenAI, which produces 1,536-dimensional embeddings for all context and query inputs. In this research, we also employed GPT-4o-mini, a compact version of GPT-4, to balance performance with computational efficiency and cost.

Another critical component is prompt engineering, which ensures that the LLM generates responses strictly based on the provided dataset, in this case, the Q&A corpus. When a user query falls outside the dataset's scope, the prompt is designed to prevent the model from generating unrelated or fabricated content. This strategy helps mitigate hallucinations and ensures that the system delivers accurate and contextually grounded responses.

F. Test Set Creation

First, we selected 14 topics that frequently appeared in the dataset. From each topic, we chose two representative questions from the database: one short-to-medium question (50–150 characters in length) and one long question (more than 200 characters in length). Using an LLM, we summarized and paraphrased the corresponding original answers to produce reference responses, which served as the ground truth for evaluating the RAG system using RAGAS metrics. This summarization process ensured that the evaluation focused solely on the core informational content, allowing for a more objective assessment of the system's response quality.

G. RAG Evaluation

For the evaluation, we utilized the RAGAS library [24], which provides standardized tools for assessing RAG systems. The evaluation metrics included faithfulness, context recall (recall@k), context precision (precision@k), and answer correctness. These metrics were selected to comprehensively evaluate both the retrieval and generation aspects of the system. Specifically, faithfulness measures the consistency of the generated response with the retrieved context, while context recall and context precision assess how effectively relevant information is retrieved from the knowledge base. Answer correctness, on the other hand, evaluates the accuracy of the final response compared to the reference answer.

To validate the performance improvements of our proposed RAG framework, we also implemented a baseline model using a naïve RAG configuration for comparison. The naïve RAG employed two retrieval methods: BM25, a traditional keyword-based search algorithm, and FAISS, a vector-based semantic search method. By comparing the advanced RAG against these baselines, we aimed to determine the individual and combined contributions of stepback prompting, hybrid search, and reranking strategies to the overall system performance.

To determine which part of our advanced RAG system was most influential, we also conducted an ablation study by evaluating the RAG system with one component removed, iterating for each of the components in our system including BM25 search, FAISS search, stepback questions, and reranking.

TABLE I. RAGAS EVALUATION METRICS

Metrics	Naive RAG		Advanced RAG
	BM25 Search	FAISS Search	
Context Precision	0.8827	0.9362	**0.9391**
Context Recall	**0.7192**	0.7103	0.7077
Faithfulness	0.8419	0.8077	**0.8754**
Answer Correctness	0.5539	0.5595	**0.6297**

IV. RESULT & DISCUSSION

Table I presents the RAGAS evaluation metrics comparing our advanced RAG framework against two naïve RAG baseline approaches, using BM25 and FAISS retrievers. The results show that the advanced RAG system we proposed achieved improvements in answer quality while maintaining competitive performance in other retrieval metrics compared to the baseline. Most notably, the advanced RAG approach achieved an Answer Correctness score of 0.6297, which is a 13% improvement over the FAISS

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baseline of 0.5595 and a 14% improvement over the BM25 baseline of 0.5539. This gain validates the effectiveness of our advanced RAG system in producing accurate, relevant responses, specifically for mental health queries.

The Context Precision scores remained high across all three approaches. Advanced RAG led with 0.9391, followed closely by F AISS with 0.9362, while BM25 scored 0.8827. This shows that each retrieval component effectively prioritized relevant context. However, FAISS's high ranking precision didn't translate to better answer correctness. This suggests that precision alone won't guarantee optimal RAG performance in domain-specific applications.

Context Recall remained somewhat steady across approaches. BM25 achieved the highest score of 0.7192, followed closely by FAISS with 0.7103 and advanced RAG with 0.7077. This slight decrease in context recall for the advanced RAG is likely due to the hybrid method's behavior of emphasizing precision and reranked relevance, which might have pruned borderline matches that would have increased coverage metrics.

For faithfulness scores, advanced RAG with a score of 0.8754 outperformed both BM25, which achieved 0.8419, and FAISS, which achieved 0.8077. The hybrid search method helps in retrieving higher-quality documents, while the reranking helps highlight the most supportive documents and play down distracting documents. This leads to the model's claims being more traceable in the retrieved text and an overall higher faithfulness score. The FAISS results might also indicate that vector-based retrieval alone could retrieve semantically similar documents, but not contextually aligned, leading the model to produce answers that might stray from the provided evidence.

Table II presents the RAGAS evaluation metrics comparing the advanced RAG to the ablation study configuration. The advanced RAG achieved the best balance of grounding and answer quality compared to the other configurations. It was able to achieve the highest faithfulness and answer correctness, while its context precision and recall were still near the top. In contrast, the other ablations were able to achieve higher raw retrieval metrics, but falls behind on response quality. These patterns suggest that the hybrid retrieval plus reranking and stepback are able to convert retrieval documents into better grounded, more accurate answers, even when pure retrieval look marginally stronger for the ablations. So, the BM25 and FAISS search methods helped with the retrieval part of the system, while the reranking and stepback helped with the answer quality and groundedness.

These results collectively suggest that the proposed advanced RAG system is able to deliver meaningful improvements to naïve RAG techniques for mental health chatbot applications. The drastic increase in Answer Correctness while maintaining a high Faithfulness score indicates that the system can provide more helpful responses without sacrificing factual grounding, which is critical when handling mental health issues. However, the modest gains in Context Precision and Recall suggests that further optimization of the retrieval pipeline can still be done to ensure a more comprehensive and precisely ranked context retrieval.

While this RAG-based assistant achieves encouraging scores on evaluation metrics such as answer correctness, it is not and should not be treated as a replacement for licensed psychologists, therapists, or other qualified clinicians. In our view, it is best understood as a supportive first-line tool, handling straightforward questions and encouraging early help-seeking, not to diagnose conditions, deliver therapy, or offer personalized medical advice. All outputs should be interpreted with human judgement, and users should be guided toward clinical pathways that ensure safety, accountability, and continuity of care.

TABLE II. ABLATION STUDY EVALUATION METRICS

Metrics	No BM25	No FAISS	No Stepback Questions	No Reranking	Advanced RAG (Full)
Context Precision	0.9427	0.9451	0.9382	**0.9551**	0.9391
Context Recall	**0.7212**	0.7103	0.7058	0.7154	0.7077
Faithfulness	0.8171	0.8706	0.8535	0.8738	**0.8754**
Answer Correctness	0.5771	0.5953	0.5897	0.5986	**0.6297**

V. CONCLUSION

In conclusion, this study introduced an advanced RAG architecture that integrates hybrid search, step-back questioning, and a reranker mechanism. The system was implemented in a Sanity-LLM, designed to deliver informative, empathetic, and safe responses. Two open-source datasets from Hugging Face were embedded and indexed using FAISS. The retrieval pipeline combined hybrid semantic-keyword search, step-back prompting, and Cohere-based reranking, while prompt engineering guided GPT-4 to generate grounded responses with minimal hallucination.

Evaluation results using RAGAS metrics demonstrate clear improvements over baseline methods. The Advanced RAG achieved the highest *Context Precision* (0.9391), *Faithfulness* (0.8754), and *Answer Correctness* (0.6297), indicating more precise retrieval and higher factual consistency. The *Ablation Study* further confirmed that each module: hybrid search, step-back questioning, and reranking contributed meaningfully to overall performance. Notably, removing reranking or step-back questions led to decreases in answer correctness and faithfulness. These findings highlight that the proposed RAG architecture effectively balances relevance, accuracy, and grounding, offering a robust foundation for trustworthy mental health dialogue systems.

Although the performance of Sanity-LLM is good, several limitations remain. First, the current evaluation relied on open datasets rather than real patient interactions, which may limit ecological validity. Second, while retrieval and generation scores are high, empathy and safety were assessed indirectly and require validation through expert and user feedback. Third, the system depends on static datasets that

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may not fully capture the diversity of real-world mental health expressions. Finally, as with all LLM-based systems, data privacy, bias, and interpretability remain critical concerns requiring deeper investigation.

For future work, expanding the knowledge base with verified clinical resources or structured knowledge graphs could enhance factual grounding. Incorporating reinforcement learning or human-in-the-loop feedback may improve personalization and long-term adaptation. Moreover, user studies with mental health professionals and end users are necessary to assess safety, trust, and therapeutic value in real-world settings.

Open Contributionship

The paper entitled "Enhancing Mental Health Chatbot with Retrieval-Augmented Generation (RAG)" was collaboratively developed by three authors under the supervision of Hidayaturrahman. Jefry Christiawan designed the RAG architecture and wrote the main sections of the manuscript. Dominick Albert Setiawan implemented the RAG system in code and collected the datasets. Joshua Andrew Pieter Oroh conducted the evaluation and wrote the result and analysis. Hidayaturrahman provided supervision, guidance, and critical feedback throughout the research.

Open Data

Data supporting this study are publicly available on HuggingFace. This research utilizes two Q&A datasets: 1. 3.51k real counseling conversation dataset that accessible on https://huggingface.co/datasets/Amod/mental_health_counseling_conversations. 2. 13.4k small Q&A mental health topic from https://huggingface.co/datasets/marmikpandya/mental-health.

Acknowledgment

The authors would like to express their gratitude to Bina Nusantara University for providing the resources, facilities, and academic support necessary for completing this research.

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